

Merge Sort

```
#include <iostream>
```

```
using namespace std;
```

```
// Function to merge two subarrays
```

```
void merge(int arr[], int left, int mid, int right) {
```

```
    int i = left;
```

```
    int j = mid + 1;
```

```
    int k = 0;
```

```
    int temp[right - left + 1];
```

```
    while (i <= mid && j <= right) {
```

```
        if (arr[i] <= arr[j]) {
```

```
            temp[k] = arr[i];
```

```
            i++;
```

```
        } else {
```

```
            temp[k] = arr[j];
```

```
            j++;
```

```
        }
```

```
        k++;
```

```
    }
```

```
// Copy remaining elements of left subarray
```

```
while (i <= mid) {
```

```
    temp[k] = arr[i];
```

```

        i++;

        k++;
    }

    // Copy remaining elements of right subarray
    while (j <= right) {
        temp[k] = arr[j];

        j++;

        k++;
    }

    // Copy sorted elements back to original array
    for (int i = left, k = 0; i <= right; i++, k++) {
        arr[i] = temp[k];
    }
}

// Merge Sort function
void mergeSort(int arr[], int left, int right) {
    if (left < right) {
        int mid = (left + right) / 2;

        mergeSort(arr, left, mid);    // Sort left half
        mergeSort(arr, mid + 1, right); // Sort right half
        merge(arr, left, mid, right); // Merge both halves
    }
}

```

```
}
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter number of elements: ";
```

```
    cin >> n;
```

```
    int arr[n];
```

```
    cout << "Enter elements:\n";
```

```
    for (int i = 0; i < n; i++) {
```

```
        cin >> arr[i];
```

```
    }
```

```
    mergeSort(arr, 0, n - 1);
```

```
    cout << "Sorted array:\n";
```

```
    for (int i = 0; i < n; i++) {
```

```
        cout << arr[i] << " ";
```

```
    }
```

```
    return 0;
```

```
}
```

Output

Enter number of elements: 3

Enter elements:

12

21

21

Sorted array:

12 21 21

=== Code Execution Successful ===